

SauceDemo Automation Test Plan & Test Suite Documentation

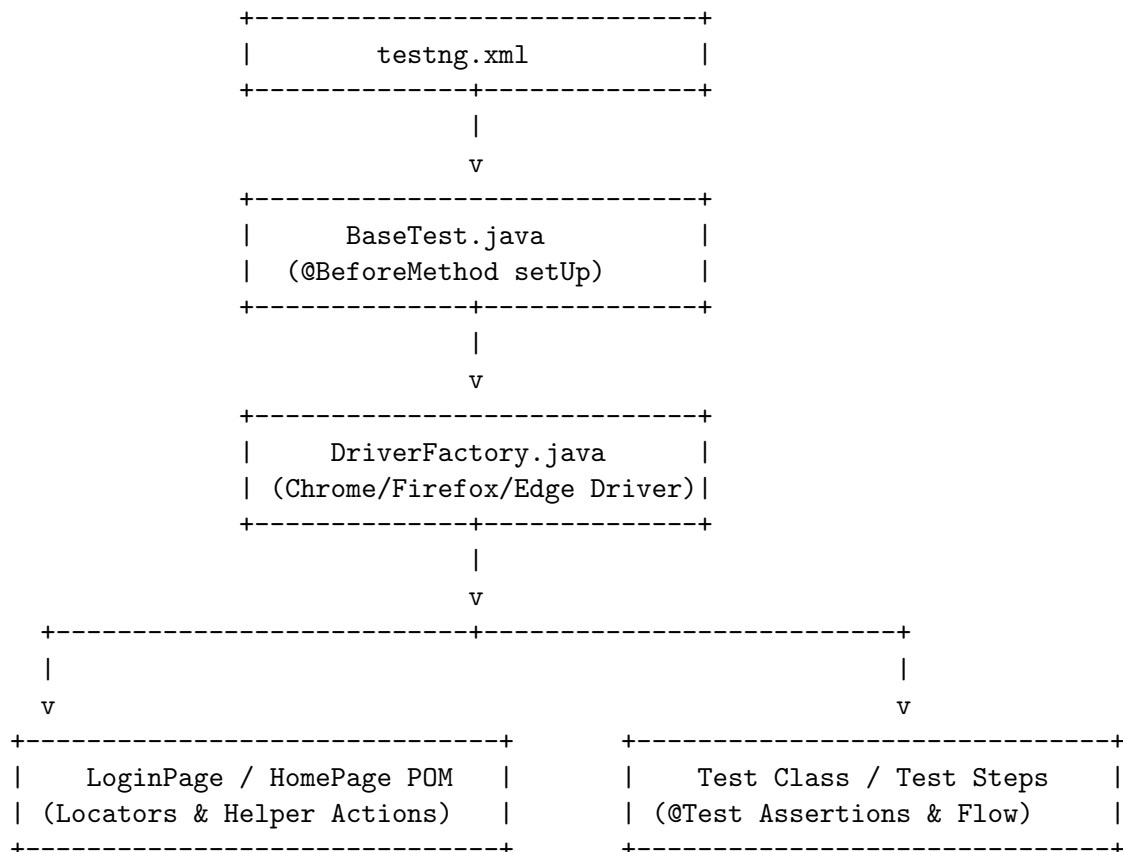
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September 27, 2026

1 Project Overview & Test Environment

- **Application Under Test (AUT):** Swag Labs (<https://www.saucedemo.com/>)
- **Test Automation Framework:** Java + Selenium WebDriver 4.x + TestNG + Maven
- **Design Pattern:** Page Object Model (POM)
- **Execution Strategy:** Cross-Browser Parallel Execution via `testng.xml`
- **Test Data:** Standard credentials (`standard_user` / `secret_sauce`)

2 Test Execution Workflow & Architecture



3 Test Cases & Step-by-Step Scenario Matrix

Test ID	Test Scenario	Execution Class	Test Steps Summary	Status
TC01-Login	User Authentication Validation	LoginTest.java	1. Enter <code>standard_user</code> 2. Enter <code>secret_sauce</code> 3. Click Login	PASS
TC02-UI	Home Page Branding Verification	HomePageTest.java	1. Wait for header 2. Assert <code>.app_logo</code> visibility	PASS
TC03-Cart	Product Selection & Cart Badge Count	CartAndSearchTest.java	1. Click Add to cart 2. Check <code>.shopping_cart_badge</code>	PASS
TC04-Sort	Product Price Sorting (Low to High)	CatalogTest.java	1. Select Price (low to high) 2. Assert price ordering	PASS
TC05-Checkout	End-to-End Purchase Flow	CheckoutTest.java	1. Proceed to checkout 2. Fill form & complete order	PASS

Table 1: Automation Test Suite Execution Summary

4 Step-Wise Detailed Test Procedures

4.1 Test Case: TC_LOGIN_001

- **Objective:** Verify user can authenticate into the application using standard user credentials.
- **Step 1 (Navigate):** Load application URL <https://www.saucedemo.com/>.
- **Step 2 (Action):** Enter text `standard_user` into element `#user-name`.
- **Step 3 (Action):** Enter text `secret_sauce` into element `#password`.
- **Step 4 (Action):** Click element `#login-button`.
- **Step 5 (Assertion):** Verify current URL contains `/inventory.html`.

4.2 Test Case: TC_CART_003

- **Objective:** Verify adding an item updates the cart item count indicator.
- **Pre-requisite:** Complete execution of TC_LOGIN_001.

- **Step 1 (Action):** Click the first available button with matching selector `button[id^='add-to-cart']`.
- **Step 2 (Assertion):** Locate `.shopping_cart_badge` element text and assert equality with `"1"`.

5 Defect Reporting & Console Log Traceability

- **SLF4J Logger Warning:** Defaulted to NOP (No-operation) due to missing logging implementation dependency in build profile.
- **CDP (Chrome DevTools Protocol) Mismatch:** Chrome version protocol mismatches do not affect standard W3C Selenium execution commands.